

Project Demonstration

Date	4 July 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyzer
Maximum Marks	5 Marks

Project Demonstration Overview

This section demonstrates the complete workflow of the **Voice-Based Concept Understanding Analyzer**. The application accepts a student's voice explanation, converts speech into text, performs semantic analysis, evaluates conceptual understanding, and generates a detailed PDF report.

1. System Architecture

DATA FLOW DIAGRAM (LEVEL 0)

Voice-Based Concept Understanding Analyzer

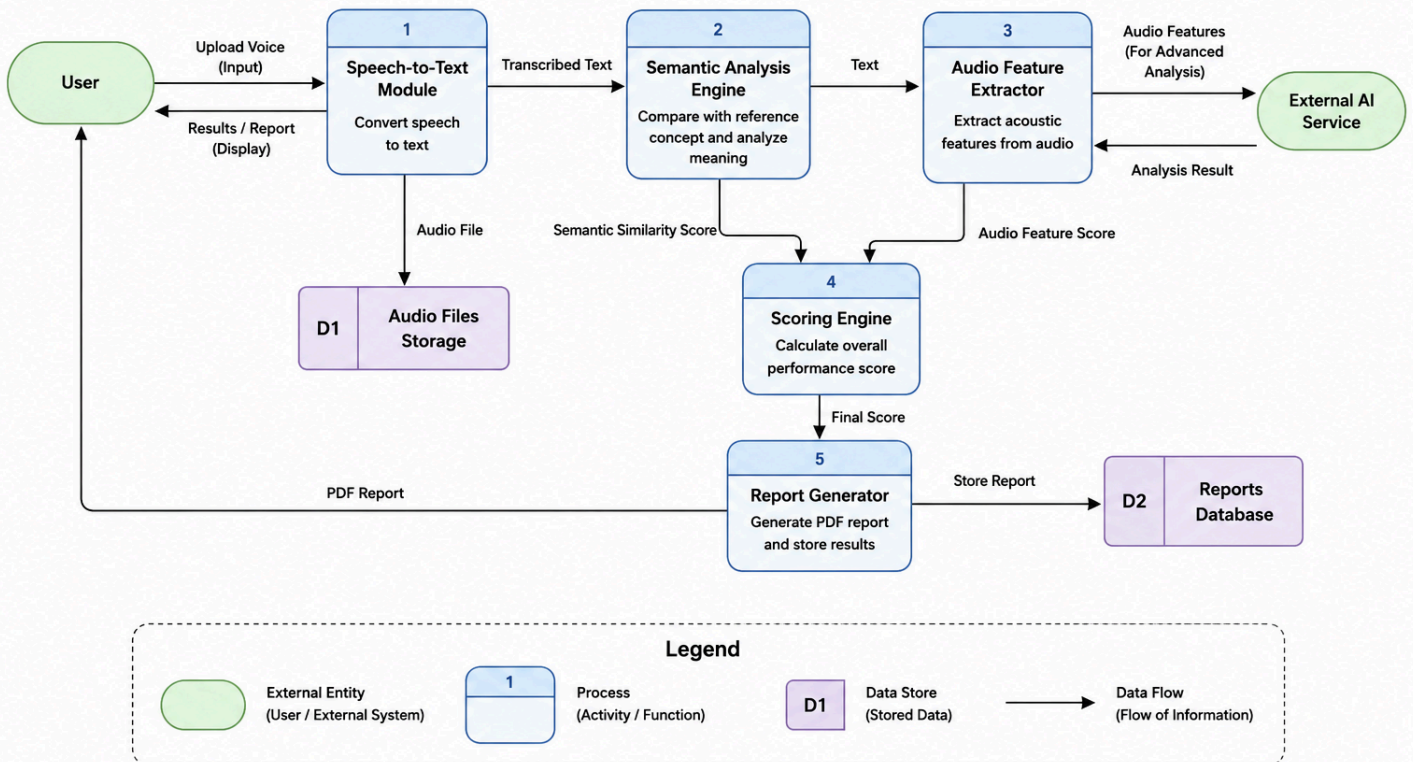


Figure 1: Data Flow Diagram

2. Project Structure

EXPLORER

...

✓ VOICE BASED CONCEPT UNDERSTANDING ANALYSIS



- > 1. Brainstorming & Ideation
- > 2. Requirement Analysis
- > 3. Project Design Phase
- > 4. Project Planning Phase
- > 5. Project Development Phase
- > 6. Project Testing
- > 7. Project Documentation
- > 8. Project Demonstration

✓ Assets

✓ Sample Audio

✓ Sample Reports

✓ Source_Code

✓ assets

 waveform.png

✓ data

 reference_concepts.json

> docs

✓ reports

 Voice_Analysis_Report.pdf

✓ src

> __pycache__

 audio_features.py

 audio_visualizer.py

 pdf_report.py

 scoring.py

M

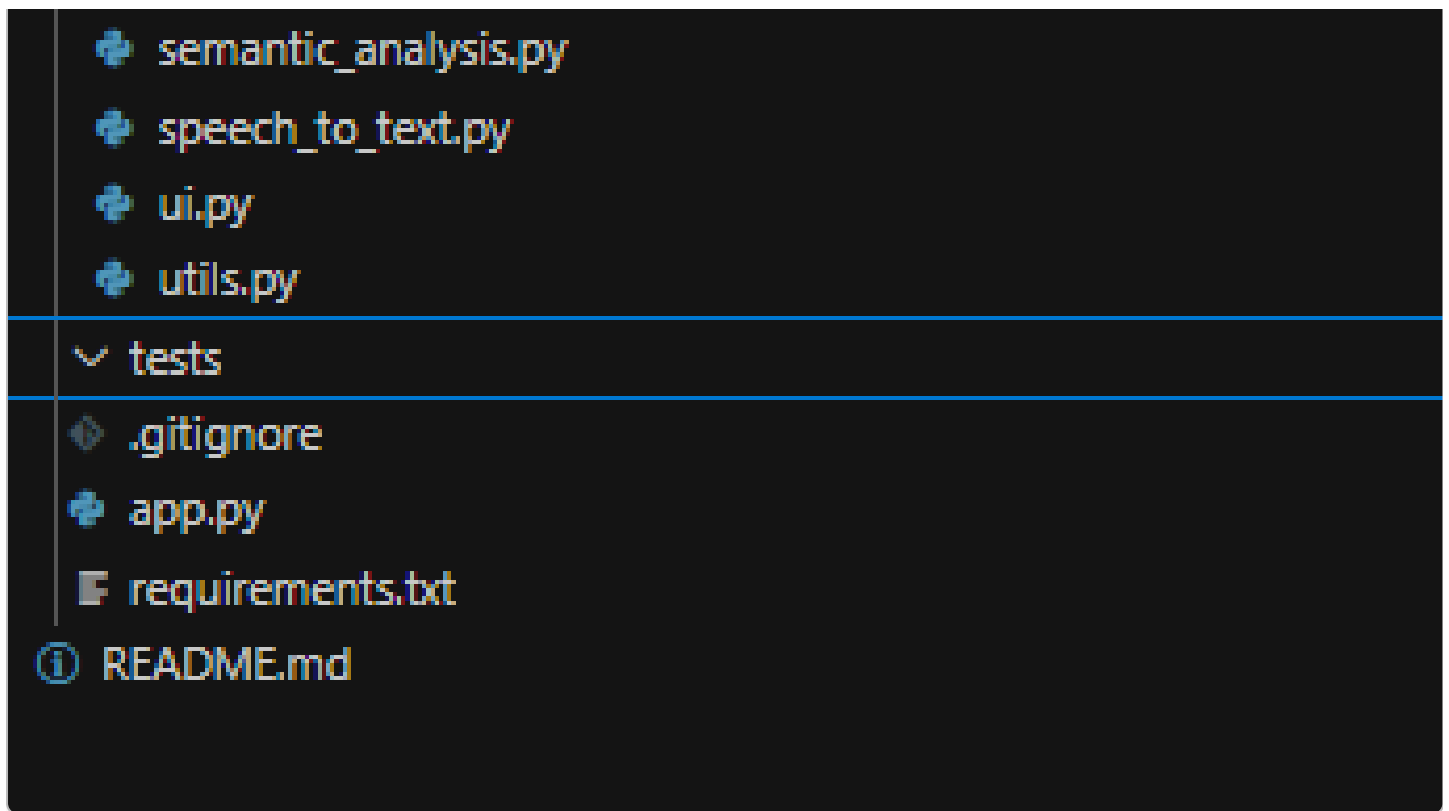


Figure 2: Project Folder Structure in VS Code

3. Running the Application

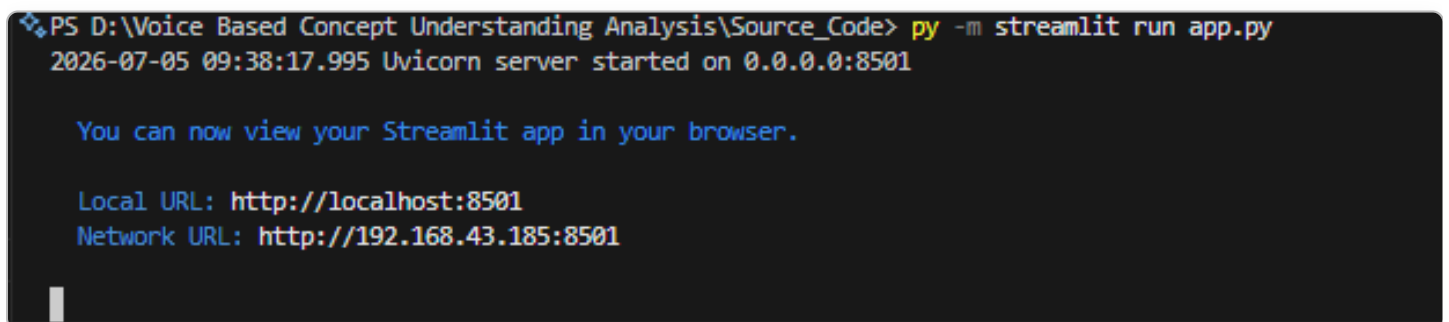


Figure 3: Streamlit Server Started Successfully

4. Streamlit Application Running

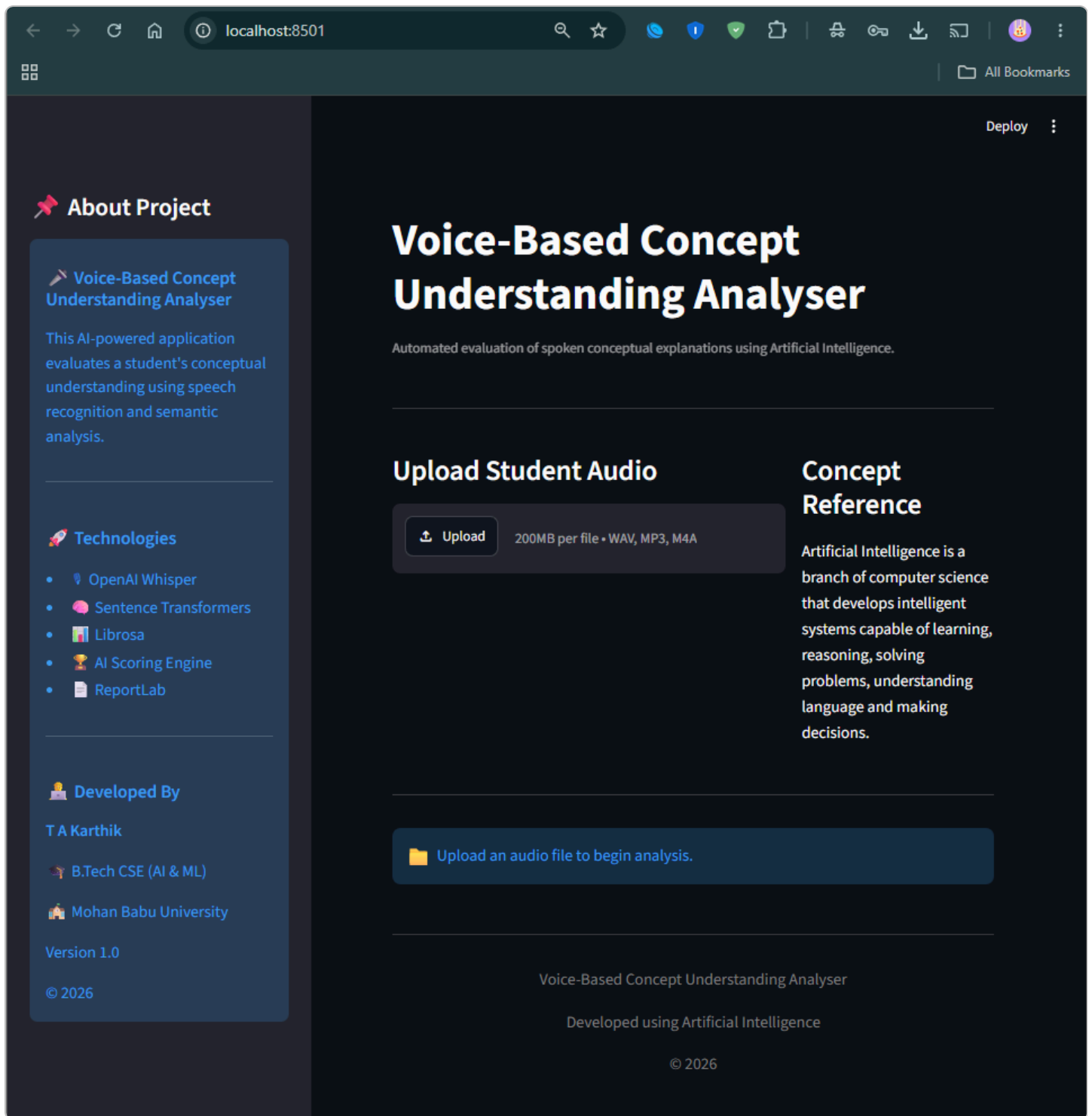


Figure 4: Application Running on Localhost

5. Application Workflow

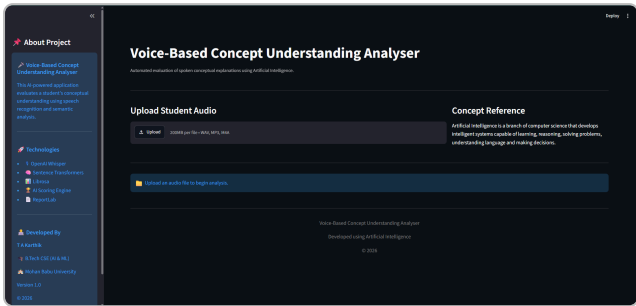


Figure 5: Home Page

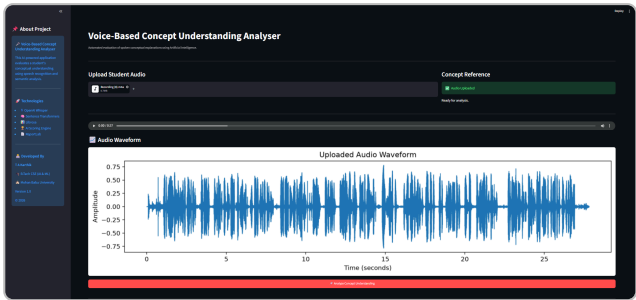


Figure 6: Audio Upload

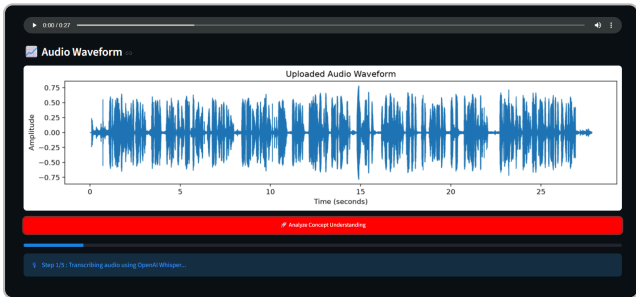


Figure 7: Audio Waveform

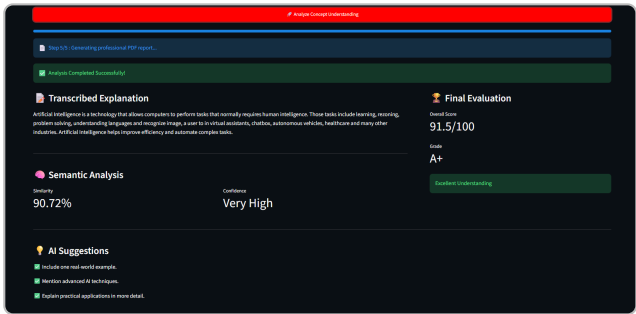


Figure 8: AI Evaluation Result

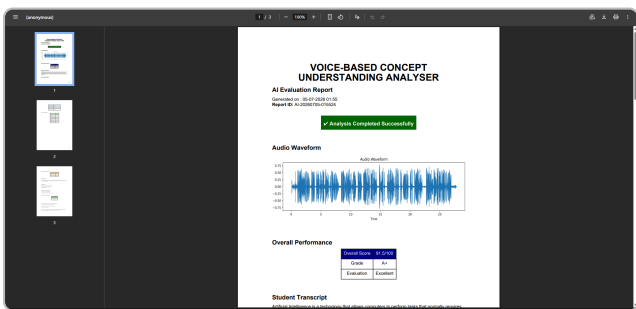


Figure 9: Generated PDF Report



Figure 10: Strength & Semantic Analysis

6. Demonstration Summary

The application was successfully executed and tested. All major functionalities including audio upload, speech-to-text conversion, semantic similarity analysis, score generation,

waveform visualization, dashboard display, and PDF report generation worked successfully during the demonstration.